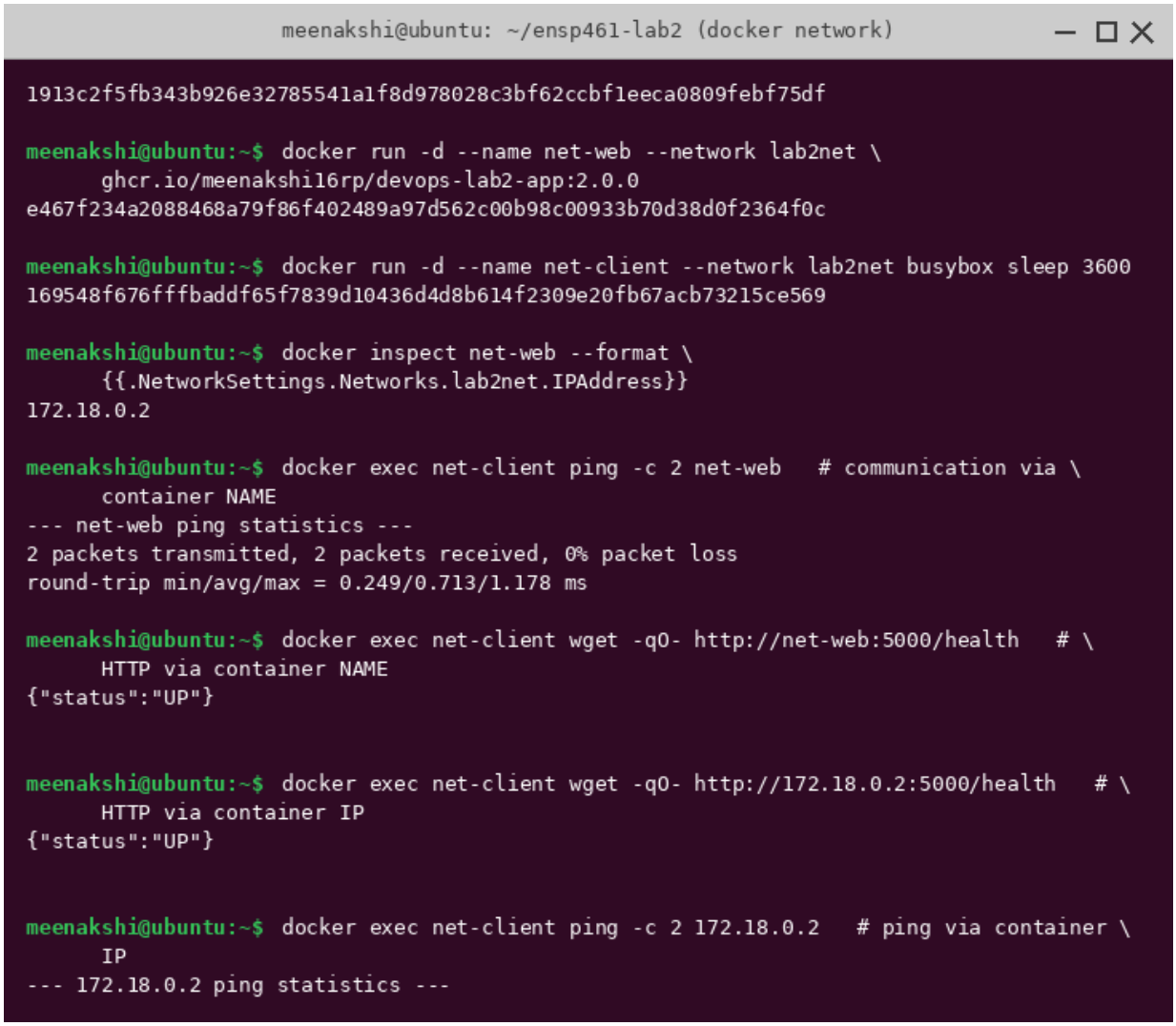
Question 4: Demonstrate communication between two containers through a Docker network, using container names and container IP addresses.

Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-2

1. User-defined bridge network with two containers



2. Observations

- On a user-defined bridge network Docker's embedded DNS resolves container names: ping net-web and wget http://net-web:5000/health both worked from net-client.
- Direct IP communication also worked (172.18.0.2) - ping and HTTP both succeeded.
- Container IPs change when containers are recreated; name-based communication is the stable, recommended pattern.
- The Flask container was reached without publishing any port - the traffic stayed inside the lab2net bridge network.